## Investigation: MUNDIAL S.A - PRODUTOS DE CONSUMO (CVM code 5312), fiscal year 2010, as it was on 2015-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Têxtil e Vestuário (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2010 DFP, version 2, filed on 2011-04-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-04-04); 2 (2011-04-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=6315

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.106 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 75 (n = 22)
- Percentile in group and year: 72 (n = 352)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.077 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 83 (n = 26)
- Percentile in group and year: 69 (n = 462)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.005 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 63 (n = 26)
- Percentile in group and year: 62 (n = 472)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.046 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 48 (n = 26)
- Percentile in group and year: 67 (n = 476)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.269 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 75 (n = 26)

Percentile in group and year: 71 (n = 473)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.837 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 20 (n = 22)
Percentile in group and year: 18 (n = 363)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.025 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 67 (n = 26)
Percentile in group and year: 62 (n = 470)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.024 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 71 (n = 26)
Percentile in group and year: 78 (n = 499)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.144 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 56 (n = 26)
Percentile in group and year: 77 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 8.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 18.7 days)
Percentile in sector and year: 83 (n = 26)
Percentile in group and year: 78 (n = 444)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 29.61 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 75 (n = 26)
Percentile in group and year: 85 (n = 440)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.024 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 67 (n = 26)
Percentile in group and year: 77 (n = 492)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −0.685 (index)
Components: X1 = −0.131; X2 = −0.021; X3 = 0.018; X4 = 0.117 (indices)
Percentile in sector and year: 30 (n = 28)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 1/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): —
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): —
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 56 accounts above the R$1,000 tolerance (group a, totals and key lines: 5; group b, reclassifications: 51), in the annex at the end.

## What was known on the date

- FCA 2010, v1: 2010-07-10
- DFP 2010, v1: 2011-04-04
- DFP 2010, v2: 2011-04-04

- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-02
- DFP 2011, v3: 2012-04-09

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da MUNDIAL S.A. - PRODUTOS DE CONSUMO em 31 de dezembro de 2010, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."

Auditor in the 2012 FCA: Directa Auditores, KPMG Auditores Independentes.

Auditor in the 2013 FCA: KPMG Auditores Independentes.

Auditor in the 2014 FCA: KPMG Auditores Independentes.

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: Directa Auditores -> KPMG Auditores Independentes (transition: two firms in the 2012 FCA). The outgoing firm audited 3 consecutive fiscal years (2009 to 2011), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-10T19:32:42+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) DRE 3.11 Lucro/Prejuízo Consolidado do Período: 10.9 in the version used, 13.0 in the comparative (difference of 2.0)
(a) BPP 2.01 Passivo Circulante: 307.0 in the version used, 308.7 in the comparative (difference of 1.7)
(a) BPP 2.02 Passivo Não Circulante: 480.9 in the version used, 479.2 in the comparative (difference of −1.7)
(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −10.1 in the version used, −8.4 in the comparative (difference of 1.7)
(a) BPP 2.03 Patrimônio Líquido Consolidado: 92.0 in the version used, 92.0 in the comparative (difference of −0.0)
(b) BPP 2.02.02.02.04 Impostos e Contribuições Sociais: 380.7 in the version used, 0.0 in the comparative (difference of −380.7)
(b) DFC (indirect method) 6.01.01.14 Imp. de Renda e Contr. Social Diferido - Passivo: 24.2 in the version used, −53.1 in the comparative (difference of −77.4)
(b) DFC (indirect method) 6.01.01.13 Atualização de Impostos e Contribuições: −53.1 in the version used, 0.0 in the comparative (difference of 53.1)
(b) BPP 2.01.02.01 Fornecedores Nacionais: 0.0 in the version used, 44.5 in the comparative (difference of 44.5)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 75.5 in the version used, 31.8 in the comparative (difference of −43.8)
(b) BPP 2.01.01.01 Obrigações Sociais: 64.2 in the version used, 20.4 in the comparative (difference of −43.8)
(b) BPP 2.01.03 Obrigações Fiscais: 0.0 in the version used, 43.8 in the comparative (difference of 43.8)
(b) BPP 2.01.03.01 Obrigações Fiscais Federais: 0.0 in the version used, 23.8 in the comparative (difference of 23.8)
(b) DFC (indirect method) 6.01.01.10 Depreciação, Amortização e Exaustão: 13.6 in the version used, 35.2 in the comparative (difference of 21.6)
(b) BPP 2.01.03.02 Obrigações Fiscais Estaduais: 0.0 in the version used, 19.8 in the comparative (difference of 19.8)
(b) DFC (indirect method) 6.02.03 Estorno Reserva de Reavaliação: 0.0 in the version used, −11.8 in the comparative (difference of −11.8)
(b) DFC (indirect method) 6.02.04 Intangiveis: −11.8 in the version used, 0.0 in the comparative (difference of 11.8)
(b) DFC (indirect method) 6.01.01.15 Resultado não Operacional: 35.2 in the version used, 24.2 in the comparative (difference of −11.0)
(b) DFC (indirect method) 6.01.01.11 Provisão para contingencias: 3.7 in the version used, 13.6 in the comparative (difference of 10.0)
(b) BPA 1.01.02.01 Aplicações Financeiras Avaliadas a Valor Justo: 5.1 in the version used, 0.0 in the comparative (difference of −5.1)
(b) BPA 1.01.02.01.02 Títulos Disponíveis para Venda: 5.1 in the version used, 0.0 in the comparative (difference of −5.1)
(b) BPA 1.01.02.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 0.0 in the version used, 5.1 in the comparative (difference of 5.1)
(b) BPA 1.01.02.02.01 Títulos Mantidos até o Vencimento: 0.0 in the version used, 5.1 in the comparative (difference of 5.1)
(b) DFC (indirect method) 6.01.02.11 Provisões para Contingencias: −3.5 in the version used, 0.0 in the comparative (difference of 3.5)
(b) DFC (indirect method) 6.01.01.01 Resultado Liquido do Exercício: 10.9 in the version used,

12.9 in the comparative (difference of 2.0)
(b) DRE 3.06 Resultado Financeiro: 16.2 in the version used, 18.3 in the comparative (difference of 2.0)
(b) DRE 3.06.02 Despesas Financeiras: −74.2 in the version used, −72.2 in the comparative (difference of 2.0)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: 31.7 in the version used, 33.7 in the comparative (difference of 2.0)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: 10.9 in the version used, 13.0 in the comparative (difference of 2.0)
(b) DRE 3.11.01 Atribuído a Sócios da Empresa Controladora: 10.9 in the version used, 12.9 in the comparative (difference of 2.0)
(b) DFC (indirect method) 6.01.01.18 Provisões Diversas Passivos: 0.2 in the version used, −1.8 in the comparative (difference of −2.0)
(b) DFC (indirect method) 6.03.03 Juros de Capital Proprio Propostos: 1.7 in the version used, −0.1 in the comparative (difference of −1.8)
(b) BPP 2.01.05 Outras Obrigações: 18.3 in the version used, 20.0 in the comparative (difference of 1.7)
(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 0.0 in the version used, 1.7 in the comparative (difference of 1.7)
(b) BPP 2.02.02 Outras Obrigações: 383.1 in the version used, 381.4 in the comparative (difference of −1.7)
(b) BPP 2.02.02.02 Outros: 383.1 in the version used, 381.4 in the comparative (difference of −1.7)
(b) BPP 2.02.02.02.05 Juros de Capital Próprio: 1.7 in the version used, 0.0 in the comparative (difference of −1.7)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: 19.1 in the version used, 17.4 in the comparative (difference of −1.7)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −9.4 in the version used, −7.7 in the comparative (difference of 1.7)
(b) BPP 2.01.05.02 Outros: 18.3 in the version used, 19.7 in the comparative (difference of 1.4)
(b) DFC (indirect method) 6.01.02.06 Imposto de Renda e Contribuição Social Diferidos: 0.0 in the version used, −1.1 in the comparative (difference of −1.1)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −45.8 in the version used, −44.8 in the comparative (difference of 1.0)
(b) BPP 2.03.01 Capital Social Realizado: 29.7 in the version used, 28.8 in the comparative (difference of −0.9)
(b) BPA 1.02.01.01 Aplicações Financeiras Avaliadas a Valor Justo: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.02.01.01.02 Títulos Disponíveis para Venda: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.02.01.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) BPA 1.02.01.02.01 Títulos Mantidos até o Vencimento: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) DFC (indirect method) 6.01.02.13 Outras Contas a Pagar: 0.4 in the version used, 0.0 in the comparative (difference of −0.4)
(b) BPP 2.01.05.01 Passivos com Partes Relacionadas: 0.0 in the version used, 0.3 in the comparative (difference of 0.3)
(b) BPP 2.01.05.02.04 Outras Contas a Pagar: 18.3 in the version used, 17.9 in the comparative (difference of −0.3)
(b) BPP 2.03.04 Reservas de Lucros: 27.3 in the version used, 26.9 in the comparative (difference of −0.3)
(b) BPP 2.03.04.10 Reserva Orçamento de Capital: 24.1 in the version used, 23.7 in the comparative (difference of −0.3)
(b) BPP 2.03.06 Ajustes de Avaliação Patrimonial: −1.5 in the version used, −1.3 in the comparative (difference of 0.2)

(b) BPP 2.01.03.03 Obrigações Fiscais Municipais: 0.0 in the version used, 0.2 in the comparative (difference of 0.2)
(b) BPP 2.01.02 Fornecedores: 44.5 in the version used, 44.5 in the comparative (difference of 0.0)
(b) DFC (indirect method) 6.01.01.12 Variação Cambial de Fornecedores: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)